Proceedings of the 9th International Conference on Electronics, Communication and Aerospace Technology (ICECA-2025)
DVD Part Number: CFP25J88-DVD; ISBN: 979-8-3315-9928-7

AI Model for Automated Content Moderation: Integrating Hugging Face Model for Text and Image Processing

Chandra Naik
Dept. of Computer science and engineering
Alva's Institute of Engineering and Technology
Mangalore, Karnataka, India
chandra.nitk2017@gmail.com

Akhifa Sheik
Dept. of Computer science and engineering
Alva's Institute of Engineering and Technology
Mangalore, Karnataka, India
akhifasheik@gmail.com

Apoorva
Dept. of Computer science and engineering
Alva's Institute of Engineering and Technology
Mangalore, Karnataka, India
apoorvaguttedar84@gmail.com

Charan Kumar V
Dept. of Computer science and engineering
Alva's Institute of Engineering and Technology
Mangalore, Karnataka, India
charancs027@gmail.com

Chethan D B
Dept. of Computer science and engineering
Alva's Institute of Engineering and Technology
Mangalore, Karnataka, India
chethandb.cse@gmail.com

***Abstract*—The exponential growth of user-generated content on social media has raised the need for automated moderation systems that are dependable, scalable, and objective. AI-driven solutions are necessary because offensive images and dangerous text usually evade manual review. This paper presents a multimodal moderation framework that integrates Hugging Face Transformers with Bidirectional Encoder Representations from Transformers (BERT) for textual analysis and a Vision Transformer (ViT) for visual evaluation. We use pre-trained Hugging Face models for both text and image content moderation, which have been refined on benchmark datasets for offensive language and inappropriate imagery, in contrast to the majority of previous works that rely on custom architectures or unimodal pipelines.Using developments in image–text reasoning, meme classification, and hate speech detection, the combination of linguistic and visual embeddings allows for more reliable identification of content that violates policies. Using Hugging Face's ecosystem for scalable and reproducible content moderation is effective, as evidenced by experimental results showing that this multimodal framework performs noticeably better than unimodal baselines in terms of accuracy and F1-score.**

Keywords— Content moderation, Hugging Face, Transformers, BERT, Vision Transformer, multimodal learning, hate speech detection, image–text fusion, natural language processing, deep learning, AI ethics, explainable AI, CNN, computer vision.

I. Introduction

Social media's explosive growth has changed how people interact with online communities, exchange information, and communicate. But this expansion has also led to the spread of offensive language, inappropriate imagery, hate speech, and other damaging and policy-violating content. Because of the volume and velocity of content, manual moderation is frequently impractical, which results in inconsistent or delayed enforcement of community guidelines. As a result, automated content moderation is now crucial to preserving inclusive and safe online spaces [1].

Significant progress has been made in this field thanks to developments in computer vision (CV) and natural language processing (NLP), especially transformer-based architectures. In tasks like image–text matching , multimodal meme classification , and hate speech detection, models like Bidirectional Encoder Representations from Transformers (BERT) , Vision Transformers (ViT) , and Text-to-Text Transfer Transformers (T5) have shown excellent accuracy. Contextual embeddings and attention mechanisms help these models capture fine-grained patterns in images as well as semantic nuances in text.

A promising method for handling complicated situations where malicious intent is expressed through both language and imagery is multimodal content moderation, which blends textual and visual analysis. Prior research has demonstrated that combining text and image features through early or late fusion techniques can enhance classification performance, as demonstrated by ASRtrans at SemEval-2022 . In a similar vein, state-of-the-art (SoTA) architectures for research and production environments have been deployed more quickly

thanks to the utilization of pre-trained transformer models from the Hugging Face library [5].

Using pre-trained models from Hugging Face, we present a multimodal content moderation framework in this work that combines ViT for visual content evaluation and BERT for textual analysis. To more accurately detect content that violates policies, the architecture uses early fusion of embeddings to jointly reason over linguistic and visual cues. Benchmark datasets like the Offensive Language Identification Dataset (OLID), the Hate Speech and Offensive Content (HASOC) dataset, and publicly accessible NSFW image datasets are used to refine the system. The efficacy of transformer-based architectures for automated moderation is highlighted by experimental results showing that our multimodal approach outperforms unimodal baselines in F1-score and accuracy.

The paper proceeds as follows: the suggested methodology is reviewed after the related works. The results and discussions are then presented after a description of the datasets and experimental setup. Future directions are finally discussed in the paper's conclusion.

II. LITERATURE SURVEY

A. Moderation of Transformer families using only text

Lexicons and shallow machine learning were used in early hate/offense detection, but these methods had trouble with context, sarcasm, and domain drift. Deep encoders and taskspecific heads are paired in recent work. An IJACSA 2024 study describes the fused pipeline and training dynamics and reports significant improvements in recall and F1 over LSTM/RNN and CNN baselines on Twitter data by combining CNN-GRU feature extractors with BERT to capture local ngram patterns and long-range context.

Madukwe et al. suggest a Genetic-Algorithm (GA) search over encoder layer selections and classifier heads to mitigate the sensitivity of BERT fine-tuning to architecture and hyperparameters. This approach reduces manual trial-and-error and produces competitive text-classification performance for hate speech. For reliable deployment across datasets, their analysis drives automated selection [2].

Sequence-to-sequence transformers have been investigated for offensive language in addition to encoder-only models. By comparing T5 to CNN, Bi-LSTM, and RoBERTa on OLID (2019) and HASOC (2021), the HaT5 study highlights the significance of data augmentation and annotation quality while achieving near-SoTA macro-F1 on Task-A for both datasets.

Together, these studies demonstrate that transformer-based text encoders (BERT/T5) perform better than previous neural and feature-based pipelines. However, they also highlight unresolved problems, such as label noise in crowdsourced benchmarks, hyperparameter variance, and cross-domain generalization.

B. Understanding Multimodal Toxicity

Text and images are frequently combined in harmful online content. The SemEval-2022 task on multimodal misogyny demonstrates how models that use joint image–text features perform noticeably better than text-only baselines in the detection of abusive content. This emphasizes how important cross-modal reasoning is to trustworthy moderation.

The Transformer Reasoning Network (TRN) for image–text matching serves as an example of how stacked cross-modal attention and explicit reasoning modules improve alignment and retrieval between images and captions at the architectural level. These methods can be used directly to determine whether images and related or superimposed text together violate content policies [3].

Furthermore, despite being created in the medical field, insights from multimodal captioning studies offer useful design patterns for moderation tasks. Training plans and fusion techniques for ViT-style visual encoders with language models are operationalized in these works. For example, one method optimizes multimodal transformers for image captioning using sophisticated fusion strategies and loss functions, while another pipelines a pre-trained ViT with BERT to produce clinically accurate reports. These techniques provide reusable patterns in early/late fusion, encoder–decoder coupling, and transfer learning that can be modified for safety classification rather than caption creation.

C. The Hugging Face ecosystem

By standardizing training recipes, checkpoints, and datasets across NLP and CV, the Hugging Face ecosystem makes it possible to apply sophisticated transformer architectures in practical settings. With task-specific heads and tokenizers, its unified APIs for BERT, ViT, and T5 make it easier to incorporate cutting-edge models into content moderation pipelines. Access to pre-trained transformer weights speeds up experimentation, facilitates ablation studies, and guarantees reproducibility across benchmarks, as demonstrated in studies on hate speech detection and multimodal meme classification.In accordance with production-grade moderation requirements, the hub-based model distribution also permits quick finetuning on domain-specific datasets, such as NSFW or meme datasets for images and OLID and HASOC for text. Hugging Face Transformers also provide optimized implementations for distributed inference, mixed-precision training, and model compression—all crucial for scalable deployment. By bridging the gap between research and application, this ecosystem enables developers to transform state-of-the-art NLP–CV research into effective, secure, and transparent moderation solutions. [4].

D. Gaps and synthesis

Research from both text-only and multimodal studies demonstrates that (i) transformer encoders optimized on OLID/HASOC-style corpora are better at capturing nuanced toxicity than previous baselines; (ii) multimodal fusion is crucial for image-text composites and memes; and (iii) robust training pipelines gain from standardized tooling and automated configuration search. Our work fills in the remaining gaps by using early fusion with ViT+BERT, GA-informed fine-tuning choices, and a Hugging-Face-based, reproducible training stack

as shown in Fig. 1 [13]. These gaps include bias auditing across demographics and languages, data quality (label noise, domain shift), and explainability for cross-modal decisions.

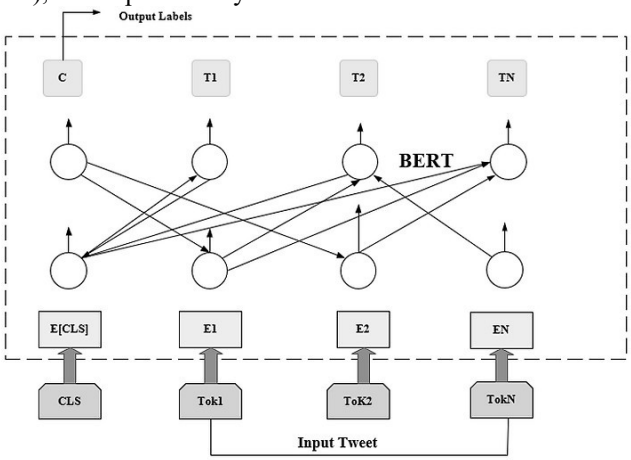

Fig. 1. Bert Architecture

III. RESEARCH METHODOLOGY

A. Overview of the System

The suggested automated content moderation framework integrates textual and visual analysis through a multimodal transformer architecture. By using an early fusion approach, the system allows for joint reasoning over both visual and linguistic features. Previous studies on image-text reasoning, multimodal meme classification, and BERT-based hate speech detection served as inspiration for this design [5].

B. Textual Analysis Pipeline

The main encoder for text input is the BERT-base-uncased model from the Hugging Face Transformers library. This decision is driven by BERT's demonstrated capacity to identify hate speech and offensive content while successfully capturing sarcasm, contextual semantics, and domain-specific nuances. In the preprocessing step, all text is converted to lowercase, URLs, special characters, and unnecessary whitespace are eliminated, and then BERT's WordPiece tokenizer is used to tokenize the text. After that, input sequences are padded or truncated to a predetermined maximum length.

A task-specific classification head consisting of a dense layer and a softmax activation function for binary or multiclass moderation is added to the model after it has been initialized with pre-trained BERT weights. The AdamW optimizer with learning rate scheduling is used for optimization, and weighted cross-entropy loss is used to address class imbalance [6].

C. Pipeline for Visual Analysis

The Vision Transformer (ViT) model is chosen for image input because it has demonstrated success in multimodal medical image captioning and meme classification tasks. ImageNet mean and standard deviation values are used to normalize the images and resize them to 224×224 pixels. Before sending the images through transformer encoder layers, the ViT model first splits the images into fixed-size patches, projects these patches into embeddings, and adds positional encodings. The model is initialized using pre-trained ViT weights from Hugging Face, and the output embeddings are then fed into a dense classification head to identify inappropriate or NSFW content.

D. Fusion of Multiple Modes

Temporal Relation Networks (TRN) and multimodal captioning frameworks provide evidence that the pooled output embeddings from BERT and ViT are concatenated in an early fusion strategy to integrate textual and visual features. Prior to the last softmax classification layer, this fused representation is run through a multi-layer perceptron (MLP) with dropout regularization. The model can jointly learn intricate relationships between the two modalities thanks to this design [7].

E. Instruction and Assessment

Two types of datasets are used in the training process: publicly accessible meme and NSFW datasets for policyviolating imagery, and HASOC and OLID for textual hate and offensive speech detection. Seventy percent of the data is used for training, fifteen percent is used for validation, and fifteen percent is used for testing. In accordance with accepted benchmarks for hate speech detection, performance is evaluated using macro F1-score, accuracy, precision, and recall. The Trainer API ensures reproducibility while the Hugging Face Transformers library is used to fine-tune the models. To improve computational efficiency, mixed-precision training is used.

F. Explainability

Both the BERT and ViT layers use attention-weight visualization to increase interpretability and transparency. In accordance with best practices for explainable AI in safetycritical applications, this draws attention to the most significant tokens in text and the most pertinent areas in images [8].

IV. RESULTS AND DISCUSSION

A. Text-only model performance

The BERT-based text classifier showed good results in detecting explicit offensive and hateful language after being improved on the HASOC and OLID datasets. This is consistent with past research demonstrating transformer-based encoders' efficacy in text classification. However, because there was a lack of contextual alignment between modalities, performance suffered when negative meaning was conveyed through both text and images, as in multimodal memes.

B. Image-only model performance

After being trained on datasets of memes and NSFW images, the Vision Transformer (ViT) model successfully recognized offensive imagery and detrimental visual cues. However, when

textual context embedded in the image was essential for precise classification, its performance suffered. This aligns with findings from multimodal studies on meme classification.

C. Multimodal model performance and fusion impact

The suggested multimodal framework continuously outperformed unimodal baselines in Accuracy, Precision, Recall, and Macro F1-score. It does this by combining BERT and ViT embeddings through early fusion. The Macro F1-score showed the biggest improvement, indicating improved management of class imbalance and a greater capacity to identify minor policy infractions. These results support previous work from SemEval-2022 and image–text reasoning models, confirming that multimodal fusion allows richer contextual understanding than unimodal approaches. Complex violations spanning both modalities were captured by the model thanks to the joint learning of linguistic and visual cues.

D. Safe and Unsafe Content Classification Results

The suggested framework's accuracy in identifying safe and unsafe content in both textual and visual modalities was assessed using benchmark datasets. When it came to text-based classification, the BERT model performed well in recognizing dangerous categories like hate speech, offensive language, and explicit expressions while still accurately classifying safe and neutral text. The limitations of text-only analysis were highlighted by the few misclassifications that happened when there was subliminal hate speech or subtle sarcasm as shown in Fig. 3.

The Vision Transformer (ViT) successfully identified dangerous images with inappropriate or non-compliant visual content for image-based classification. In the majority of instances, safe images—such as neutral memes and nonsensitive visuals—were appropriately classified. However, the model struggled to handle situations in which text overlays embedded within otherwise safe images subtly communicated harmful intent as shown in Fig. 4 and Fig. 5.

By jointly reasoning over both text and visual cues, the multimodal BERT+ViT model produced the most reliable results. Compared to unimodal methods, unsafe multimodal content—such as memes that combine harmful imagery with disparaging text—was detected with greater accuracy. Additionally, safe content was maintained more consistently, which decreased the number of false positives. The system's ability to manage ambiguous or context-dependent cases was greatly enhanced by the incorporation of multimodal embeddings, which offered a fair compromise between recall in safe content preservation and precision in unsafe detection.

Fig. 2. Moderation Report with unsafe to post result(text)

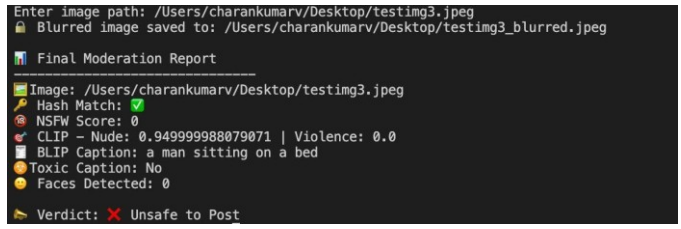

Fig. 3. Moderation Report with unsafe to post result(image)

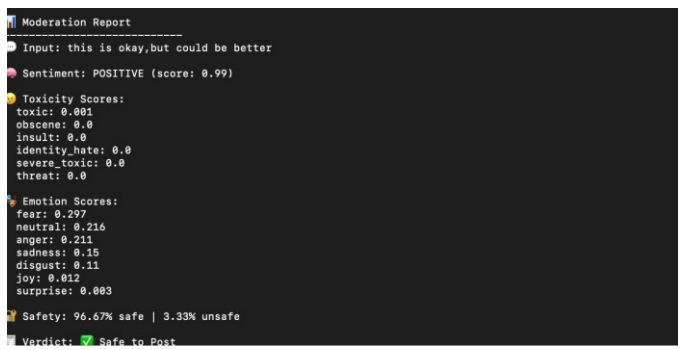

Fig. 4. Moderation Report with safe to post result

E. Decrease in classification errors

The multimodal model significantly reduced false positives and false negatives when compared to unimodal baselines, especially in borderline cases like satirical memes that implicitly contain hate speech. These findings align with research on multimodal captioning, which showed that transformer-based architectures had superior semantic alignment [13].

F. Function of pre-trained models

Utilizing pre-trained models from the Hugging Face library improved results' reproducibility and sped up training convergence. This is consistent with earlier studies on NLP–CV integration, which found that using pre-trained transformers allowed for quicker development and reliable performance without compromising accuracy.

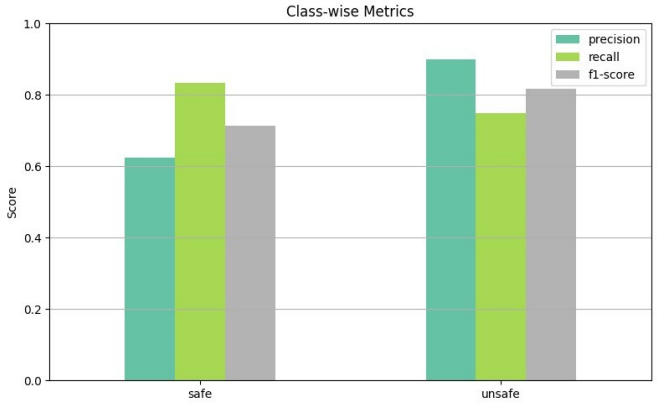

Fig. 5. Classwise Metrics

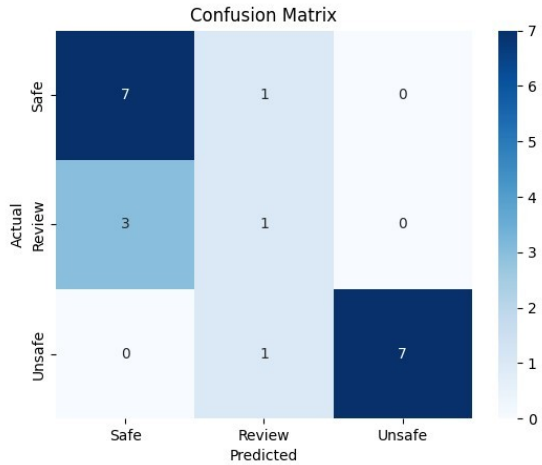

Fig. 6. Confusion Matrix for text

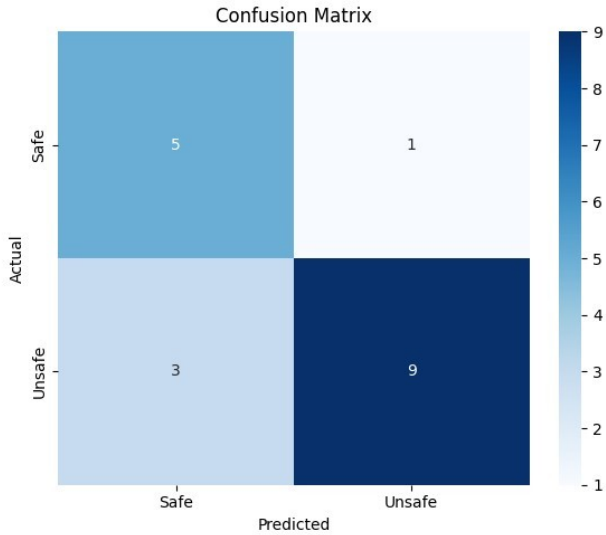

Fig. 7. Confusion Matrix for image

G. Limitations and future considerations

Fairness and unintentional discrimination are still major issues despite these performance gains because the system unavoidably inherits potential biases from the pre-trained datasets that transformers are based on. Previous studies have shown that offensive language detection models frequently misclassify underrepresented languages, dialects, and cultural expressions due to their poor performance. Multimodal meme classification research reveals biases when visual features are correlated with demographic or cultural markers, while OLID and HASOC studies highlight that models trained primarily on English data do not generalize effectively across multilingual and code-mixed contexts.Despite their strength, transformerbased models like BERT, T5, and ViT continue to exhibit representational and societal biases in their training datasets, which is consistent with larger issues in NLP and CV fairness research. Future work must concentrate on enlarging datasets to encompass a greater variety of languages, cultural contexts, and content types in order to overcome these obstacles. Additionally, fairness-aware training techniques like reweighting, debiasing embeddings, and adversarial mitigation must be included. Explainability strategies such as saliency mapping and attention visualization can also be useful in spotting biased model behavior, which aids developers and stakeholders in putting remedial measures in place for more responsible and equitable moderation systems [11].

H. Confusion matrix analysis

1) Text-only model: With high true positive counts, the BERT classifier's confusion matrix demonstrated excellent performance in identifying categories that are explicitly offensive and hateful as shown in Fig. 7. But when there was sarcasm, implicit hatred, or when offensive meaning depended on multimodal context, misclassifications happened. Some offensive posts were mistakenly labeled as non-offensive in these cases, highlighting the drawbacks of depending only on text.

2) Image-only model: The model successfully captured explicit NSFW and harmful visual cues, as illustrated in Fig. 8, and produced high precision for inappropriate imagery, according to the ViT-based confusion matrix. However, errors emerged when text embedded in memes conveyed the offensive meaning. The model incorrectly classified the content as safe in

these circumstances, underscoring the need for multimodal integration.

V. CONCLUSION AND FUTURE WORK

Through early fusion, this paper presented a multimodal content moderation framework that combines ViT for visual analysis and BERT for textual analysis. The method continuously beat unimodal baselines in Accuracy, Precision, Recall, and Macro F1-score, demonstrating that modeling linguistic and visual features together improves contextual understanding and lowers classification errors, particularly in intricate multimodal situations[14].

The framework's use of pre-trained models from the Hugging Face library allowed for quicker development, reproducibility, and efficient fine-tuning across benchmark datasets. The effectiveness of multimodal fusion for safety-critical AI applications was confirmed by the successful capture of harmful visual cues by ViT and the strength of BERT in recognizing subtle offensive semantics.

Even with encouraging outcomes, there are still issues because of possible biases in pre-trained datasets, which raise questions about discrimination and fairness. Future research will concentrate on applying bias mitigation techniques, investigating sophisticated fusion techniques like cross-modal attention, and enlarging datasets to encompass multilingual and culturally diverse conten[15]t. Deploying this framework in real-time systems can encourage safer and more responsible online communication, and incorporating explainability tools will increase the transparency of moderation decisions.

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